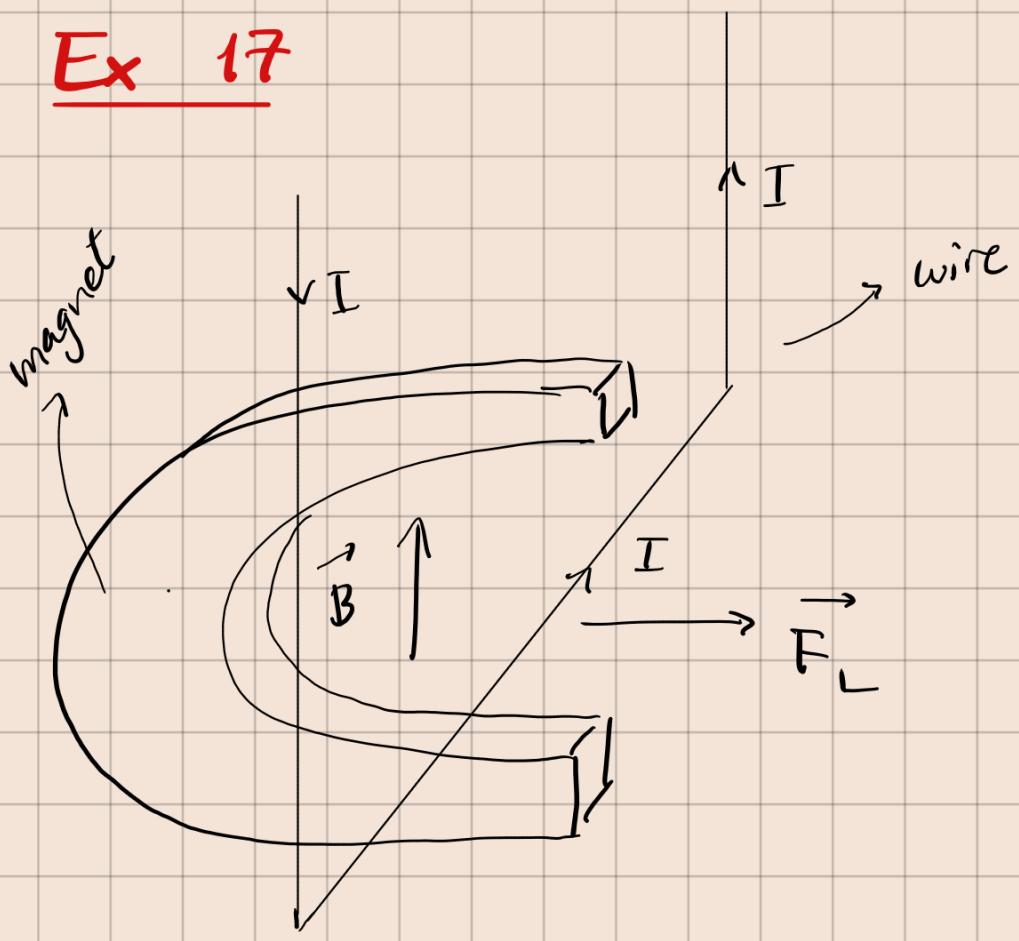
Ex 17

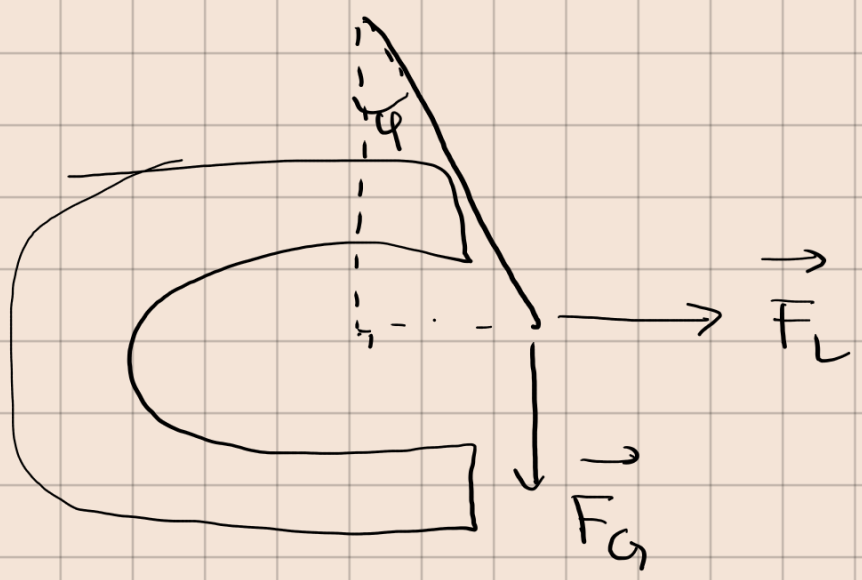


$j = I/A$

Lorentz force: $F_L = I \cdot l \cdot B = jAlB$

Gravity: $F_G = mg = \rho A l g$

Equilibrium:



$$\tan\varphi = \frac{F_L}{F_G} = \frac{jAlB}{\rho A l g} = \frac{jB}{\rho g} = 0.302$$

$$\Rightarrow \varphi = 16.8^\circ$$

Ex 18

a) Initially: ion at rest.

Acceleration: energy due to potential U is converted to kinetic energy.

$$\Rightarrow \frac{1}{2} m_1 v^2 = eU$$

$$\Rightarrow v = \sqrt{\frac{2eU}{m_1}}.$$

The ion enters the magnetic field with velocity v, and gets deflected by the Lorentz force.

Lorentz force = centrifugal force.

$$\Rightarrow \quad evB = \frac{m_1 v^2}{r_1}$$

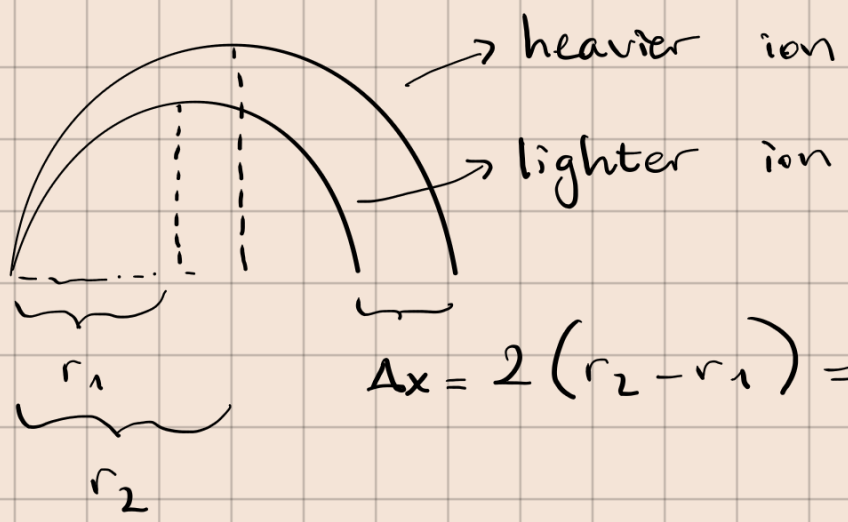
$$\Rightarrow \quad B = \frac{m_1 v}{e r_1} = \frac{m_1}{e r_1}\sqrt{\frac{2eU}{m_1}}$$

$$= \frac{\sqrt{2 m_1 U / e}}{r_1} = 0.18\ T$$

b) Use the last equation for the two ions (same mag. field):

$$B = \frac{\sqrt{2 m_2 U / e}}{r_2} = \frac{\sqrt{2 m_1 U / e}}{r_1}$$

$$\Rightarrow \quad r_2 = r_1 \sqrt{\frac{m_2}{m_1}} = 322.2\ mm$$



$$\Delta x = 2(r_2 - r_1) = 10\ mm$$

Ex 19

a) See the formula in $\boxed{307-11}$

$$B = \frac{\mu_0 N I}{\ell} \overset{!}{=} B_H$$

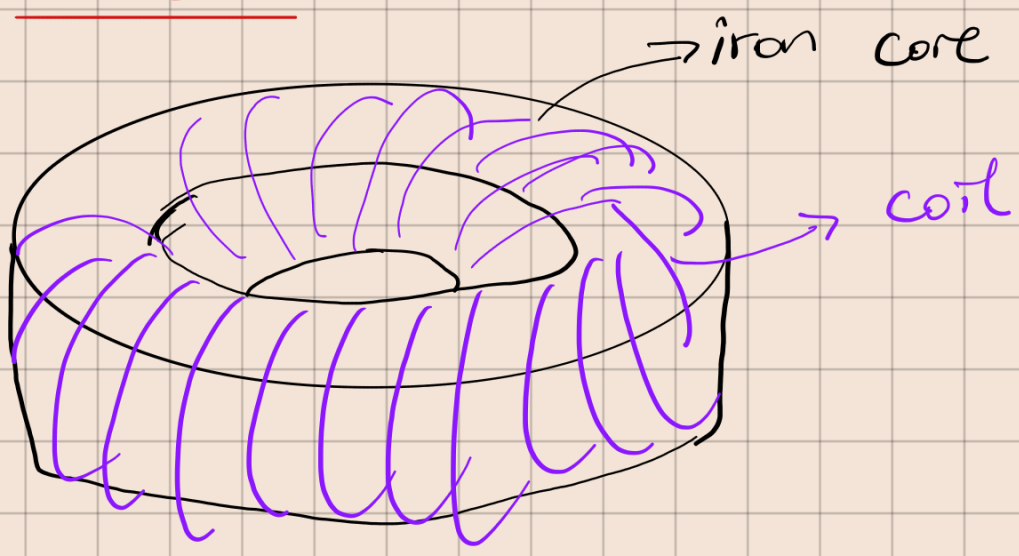
$$\Rightarrow I = \frac{B_H \cdot \ell}{\mu_0 N} = 0.33 \text{ A}$$

$$\left(\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m} / \text{A} \right)$$

b) B_H points to left, so B must point to right to compensate.

Right-hand rule $\Rightarrow I_+$.

Ex 20



See $\boxed{308-5}$

Initially: U_0, I_0 $\longrightarrow$ B-field in the iron core

After switch-off: B fie

$$U_{ind} = -L \frac{dI}{dt} \quad \text{(from script)}$$

and $U_{ind} = R \cdot I$

$$\Rightarrow \quad -L \frac{dI}{dt} = RI$$

$$\Rightarrow \quad \frac{dI}{I} = -\frac{R}{L} dt$$

Integrate both sides:

$$\Rightarrow \int_{I_0}^{I} \frac{1}{I}\, dI = -\int_{0}^{t} \frac{R}{L}\, dt$$

with $I_0 = I(t=0)$

$$\Rightarrow \ln(I) - \ln(I_0) = -\frac{R}{L} t$$

$$\Rightarrow \ln(I/I_0) = -\frac{R}{L} t$$

$$\Rightarrow I/I_0 = \exp(-R/L\, t)$$

$$\Rightarrow I = I_0 \exp(-R/L\, t) = I(t)$$

Here, use $R = U_0 / I_0 = 38\,\Omega$

and $L = \mu_r \mu_0 N^2 A / \ell$

$$= \mu_r \mu_0 N^2 (\pi d_E^2 / 4) / (\pi d_s)$$

$$= 67.9 \text{ mH}$$

plug into $I(t)$ and find

$I(t = 1 \times 10^{-3}\,s) = 2\ A.$